Übungsblatt 05: Docker & Kubernetes for Scalable Deployment

DevOps for Cyber-Physical Systems (HS 2026)

Abgabe: **Mo. 24.11.2025, 16:15 Uhr**

- **Für diese Übung erfolgt die Abgabe über GitHub:**
  - Forken Sie das Repository (Link siehe unten)
  - Bearbeiten Sie die Aufgaben in Ihrem Fork
  - Dokumentieren Sie Ihre Lösungen im `README.md`
  - Fügen Sie den Link zu Ihrem geforkten Repository im README hinzu
- Laden Sie die **`README.md`-Datei** über **ILIAS** hoch.
- Bei Abgabeproblemen in ILIAS schicken Sie die README-Datei ***rechtzeitig*** per Mail an **prakash.aryan@unibe.ch**.
- **Die Bearbeitung dieses Übungsblatts ist obligatorisch, wird jedoch nicht benotet.**

Prerequisites:

- Install **Docker Desktop**[1]
- Install **Minikube**[2] (or enable Kubernetes in Docker Desktop)
- **Fork** the lecture repository to your GitHub account: `https://github.com/prakash-aryan/lecture5-dockerk8s-demo`
- Clone **your forked repository** locally
- Run `docker compose up` and verify the app works at `http://localhost:5000`
- Create a free Docker Hub account[3]

Aufgabe 1: Modify the Docker Setup

(a) **Add Adminer Service**

Add **Adminer**[4] (a database management UI) to `docker-compose.yml`.

Requirements:

- Use the official `adminer:latest` image
- Be accessible on port `8080`
- Connect to the same network (`app-network`) as other services
- Add `depends_on` to ensure database is ready first
- Set `ADMINER_DEFAULT_SERVER=db` environment variable

Testing:

1. Run `docker compose up -d`

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[1] `https://www.docker.com/products/docker-desktop/`
[2] `https://minikube.sigs.k8s.io/docs/start/`
[3] `https://hub.docker.com/`
[4] `https://hub.docker.com/_/adminer`

2. Open `http://localhost:8080`
3. Login with these credentials (from `docker-compose.yml`):
   - System: `PostgreSQL`
   - Server: `db`
   - Username: `taskuser`
   - Password: `taskpass`
   - Database: `taskdb`
4. Take a screenshot showing the tasks table

Hint: Look at how other services (web, db, redis) are structured in `docker-compose.yml` for reference.

(b) **Change the Base Image**

Modify the `Dockerfile` to use `python:3.11-alpine` instead of `python:3.11-slim`.

a) What changes are needed to make it work with Alpine?
b) Build both versions and compare sizes using `docker images`
c) Document the size difference and any build issues you encountered

Aufgabe 2: Docker Operations

(a) **Image Tagging and Registry**

a) Build the image with a version tag: `docker build -t task-app:v1.0 .`
b) Create a Docker Hub account (free) and push your image
c) Document the commands you used and include a screenshot of your image on Docker Hub

(b) **Container Inspection**

With the app running, use these commands and briefly explain what each shows:

```
docker compose logs web
docker inspect lecture5-web
docker stats
```

Aufgabe 3: Deploy to Kubernetes

(a) **Deploy the Application**

a) Start Minikube: `minikube start`
b) Deploy: `kubectl apply -f k8s-backend.yaml` and `kubectl apply -f k8s-web.yaml`
c) Build your image: `docker build -t YOUR-USERNAME/lecture5-webapp:v1.0 .`
d) Push to Docker Hub and update `k8s-web.yaml` with your username
e) Access app: `minikube service lecture5-web-service`
f) Screenshot: app in browser with tasks + `kubectl get pods`

(b) **Scale and Test Load Balancing**

a) Scale to 5 replicas: `kubectl scale deployment lecture5-web -replicas=5`
b) Run: `python test_load_balancing.py`
c) Screenshot: load balancing test output showing traffic distribution
d) Explain: How does Kubernetes distribute traffic? (2-3 sentences)

(c) **Self-Healing**

a) Delete a pod: `kubectl delete pod POD-NAME`
b) Watch: `kubectl get pods` (run multiple times)
c) Screenshot: pods before/during/after deletion
d) Explain: Why is self-healing important? (2-3 sentences)

Submission:

1. Fork the repository: `https://github.com/prakash-aryan/lecture5-dockerk8s-demo`

2. Complete all tasks in your fork
3. Update the `README.md` in your fork to include:
   - Link to your forked GitHub repository at the top
   - Screenshots of all required outputs (embedded in README)
   - What you changed/added for each task
   - Your Docker Hub username
   - Answers to all questions (e.g., how K8s distributes traffic, why self-healing matters)
   - Any issues encountered and how you solved them
4. Commit and push all changes to your fork
5. Submit via **ILIAS**: **Upload your `README.md` file**

Resources:

Official Documentation:

- Docker Documentation: `https://docs.docker.com/`
- Docker Compose Guide: `https://docs.docker.com/compose/`
- Kubernetes Basics: `https://kubernetes.io/docs/tutorials/kubernetes-basics/`
- kubectl Cheat Sheet: `https://kubernetes.io/docs/reference/kubectl/cheatsheet/`
- Minikube Guide: `https://minikube.sigs.k8s.io/docs/`

Interactive Learning Platforms (Free):

- Play with Docker: `https://labs.play-with-docker.com/`
- Play with Kubernetes: `https://labs.play-with-k8s.com/`
- Killercoda (Docker & Kubernetes Labs): `https://killercoda.com/`
- Docker 101 Tutorial: `https://www.docker.com/101-tutorial/`

Best Practices & Tutorials:

- Docker Compose Best Practices: `https://docs.docker.com/compose/how-tos/environment-variables/best-practices/`
- Docker for Beginners (Play with Docker): `https://training.play-with-docker.com/beginner-linux/`
- Getting Started with Docker & Kubernetes: `https://www.educative.io/blog/docker-kubernetes-beginners-guide`
- Docker Roadmap: `https://roadmap.sh/docker`